



Computer Networks Week # 14

**M.Nadeem Akhtar
CS & IT Department
The University of Lahore**

Email: nadeem.akhtar@cs.uol.edu.pk

URL- <https://sites.google.com/view/nadeemuolcscn/home/course-material>

ACKNOWLEDGMENTS

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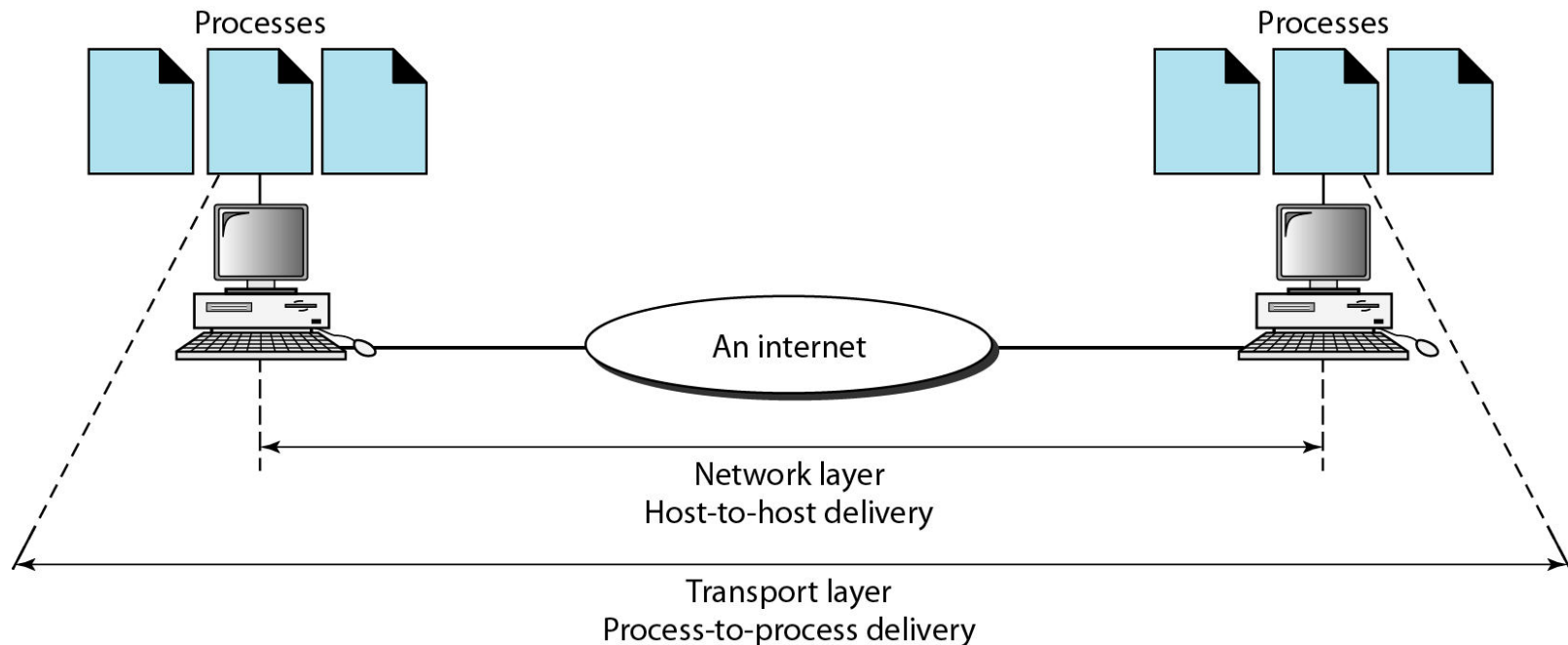
Week 14- Course Plan

Transport Layer Protocols Overview:

- User Datagram Protocol (UDP)
- Transmission Control Protocol (TCP)
- Stream Control Transmission Protocol (SCTP)

TRANSPORT LAYER IN TCP/IP MODEL

- **Transport layer** is responsible for **giving services** to the **application layer**: to **get a message** from an application program running on the **source host** and **deliver it** to the corresponding application program on the **destination host**.
- **Transport Layer performs reliable “process-to-process” delivery of a message**



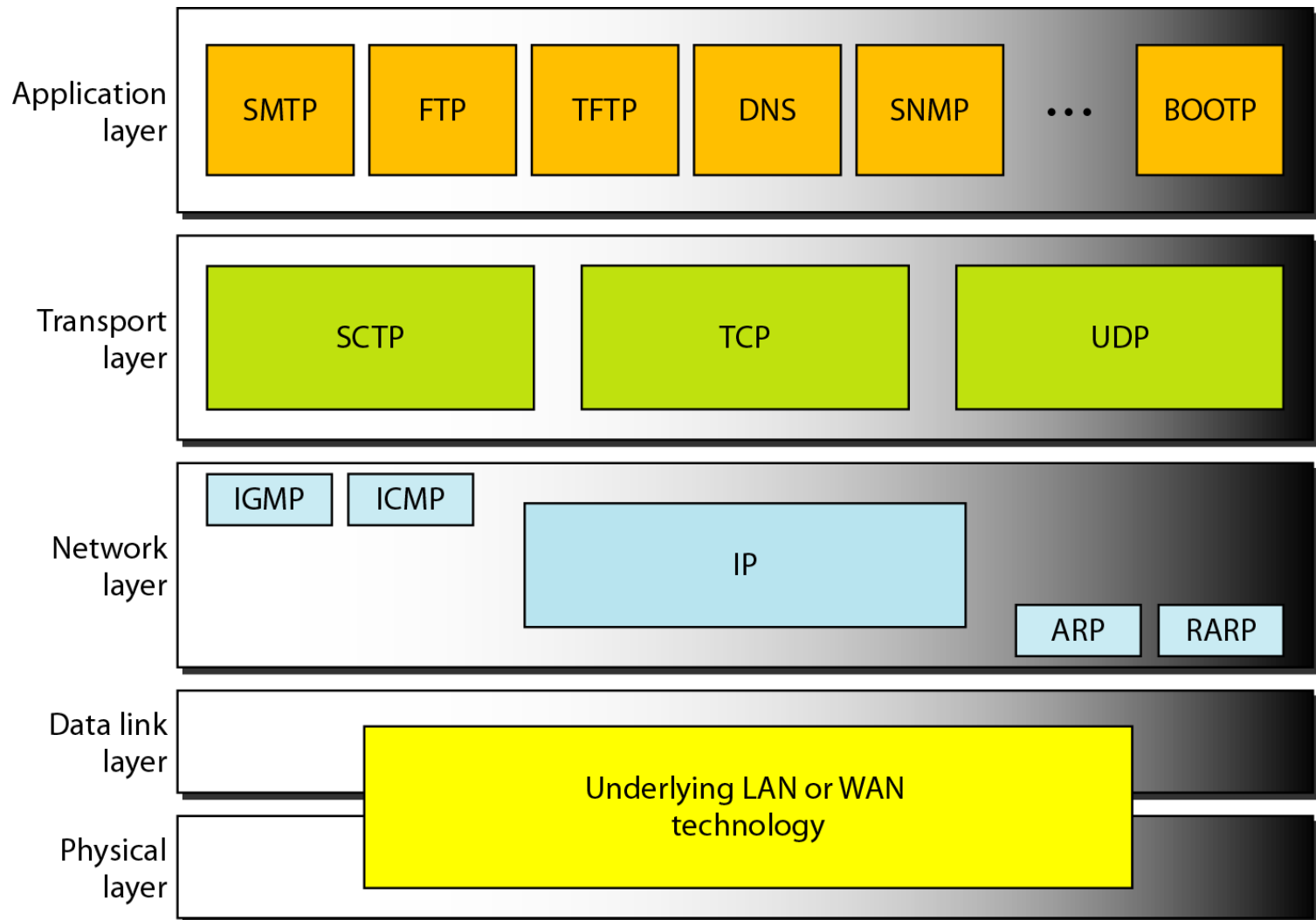
Transport Layer Services

- transport protocols run in end systems
 - **sender side:** breaks app messages into segments, passes to network layer
 - **receiver side:** reassembles segments into messages, passes to application layer

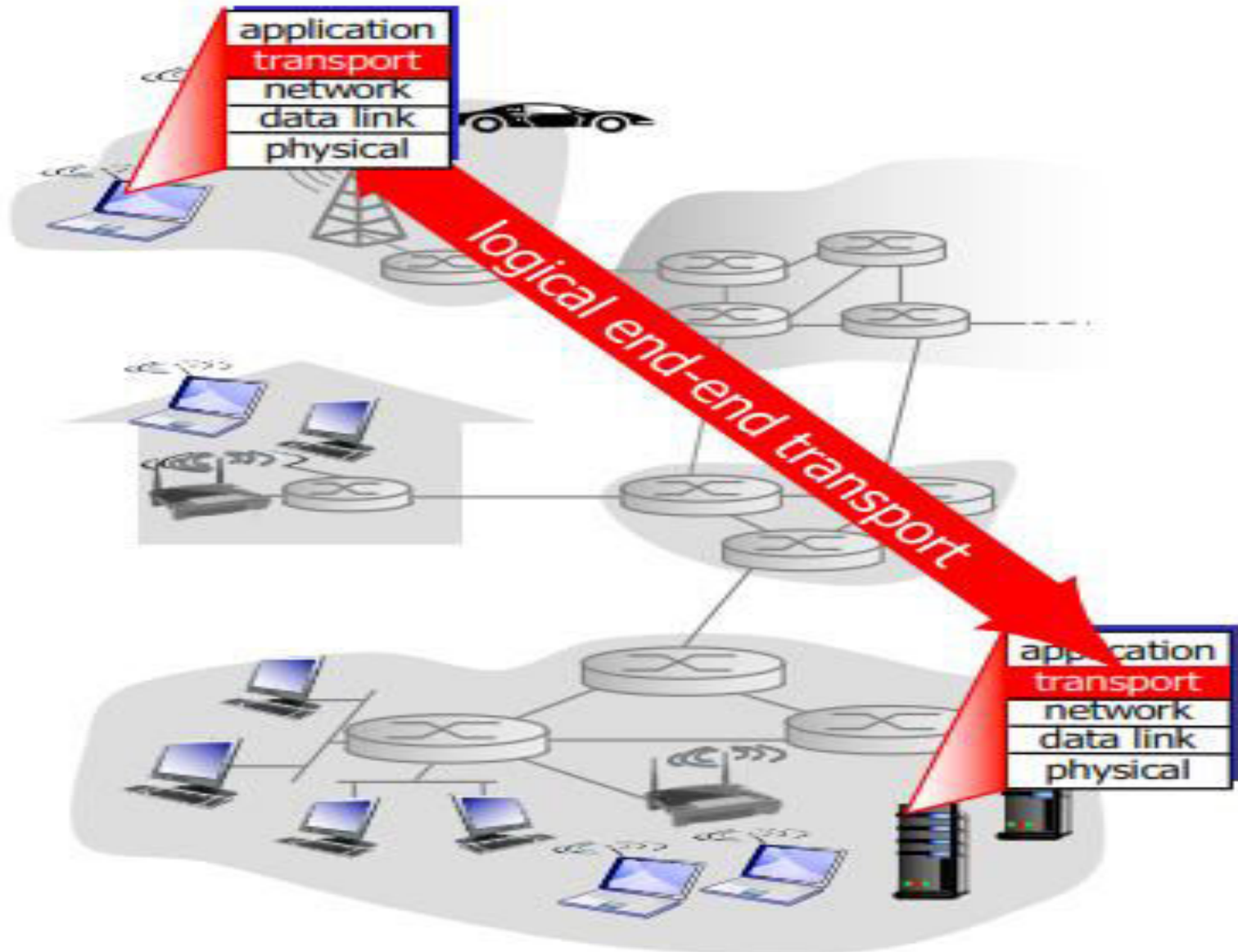
Overview of Transport layer Protocols:

- 1) **User Datagram Protocol (UDP)**, is a **connectionless** protocol that transmits user datagrams without first creating a logical connection.
Connectionless means each user datagram is independent without being related to previous or next one
- 2) **Transmission Control Protocol (TCP)**, is a **connection-oriented** protocol that first establishes a logical connection between transport layers at two hosts before transferring data.
 - a) **Connection-oriented** means a connection must be established between both ends of a transmission before either can transmit data.
 - b) Provides flow control, error control, congestion control
- 3) **Stream Control Transmission Protocol (SCTP)** is designed to respond to new applications that are emerging in the multimedia.

Position of UDP, TCP, and SCTP in TCP/IP suite



Transport Layer Services and Protocols



Transport Vs Network

network layer:

communication between hosts

transport layer:

Logical communication between processes

- relies on, enhances, network layer services

household analogy:

12 kids in Ann's house sending letters to 12 kids in Bill's house:

- hosts = houses
- processes = kids
- app messages = letters in envelopes
- transport protocol = Ann and Bill who demux to in-house siblings
- network-layer protocol = postal service

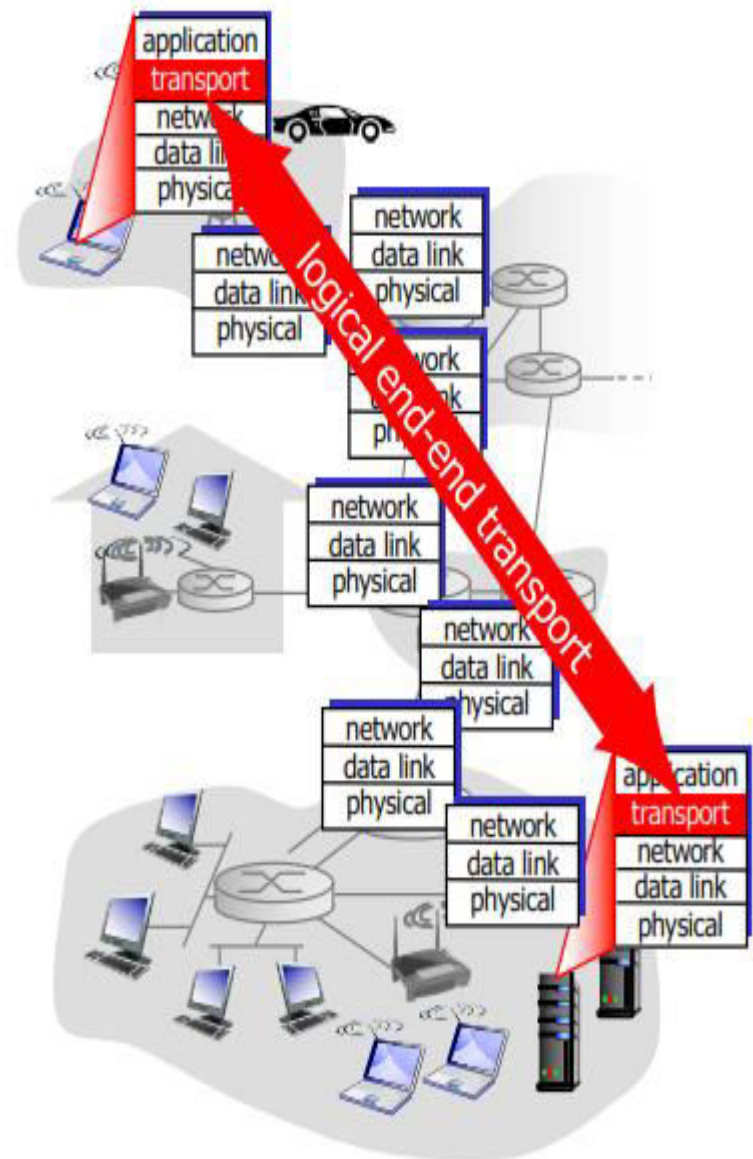
Internet Transport Layer Protocols

Reliable, in-order delivery (TCP)

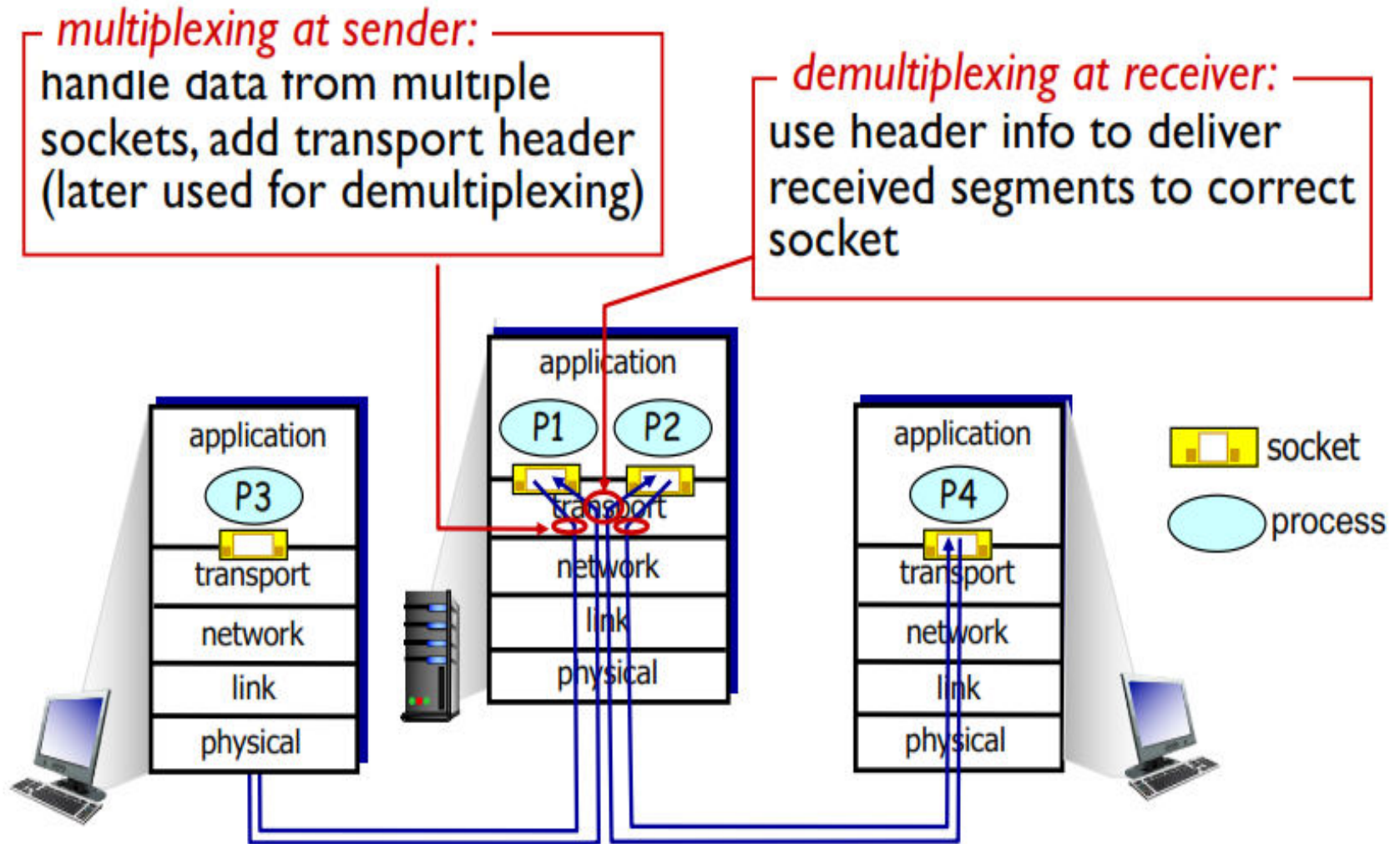
- congestion control
- flow control
- connection setup

unreliable, unordered delivery: UDP

- extension of
best-effort IP



Multiplexing/Demultiplexing

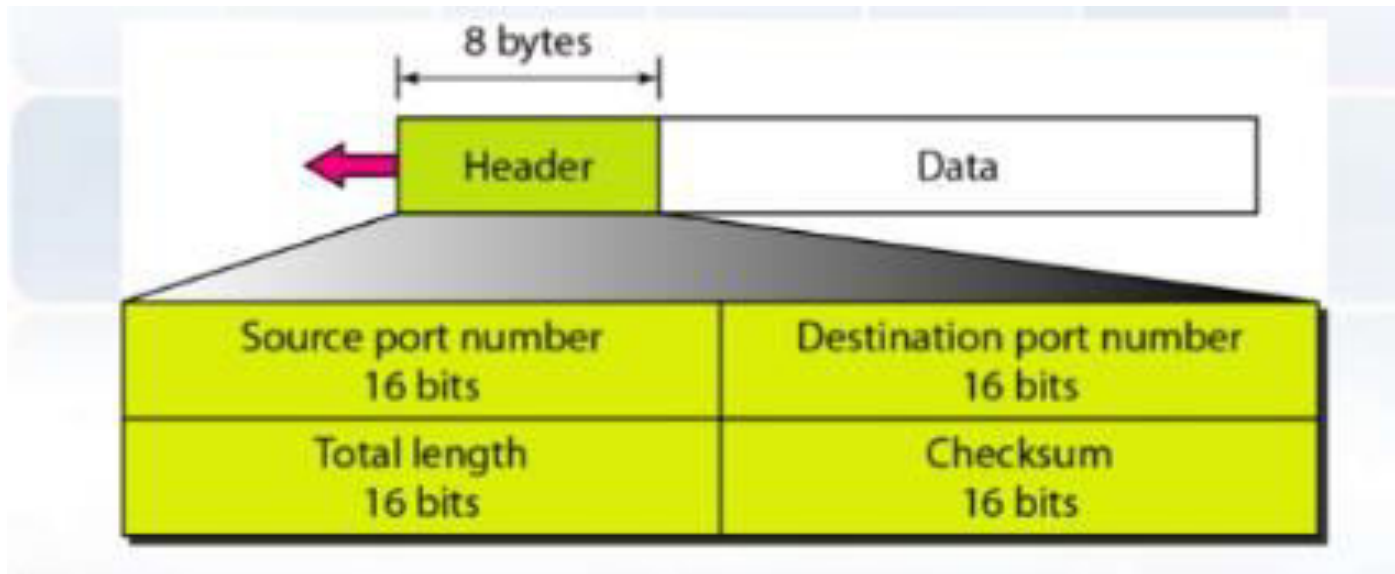


Transport layer creates a unique connection via socket (combination of IP and port number)

1-USER DATAGRAM PROTOCOL (UDP)

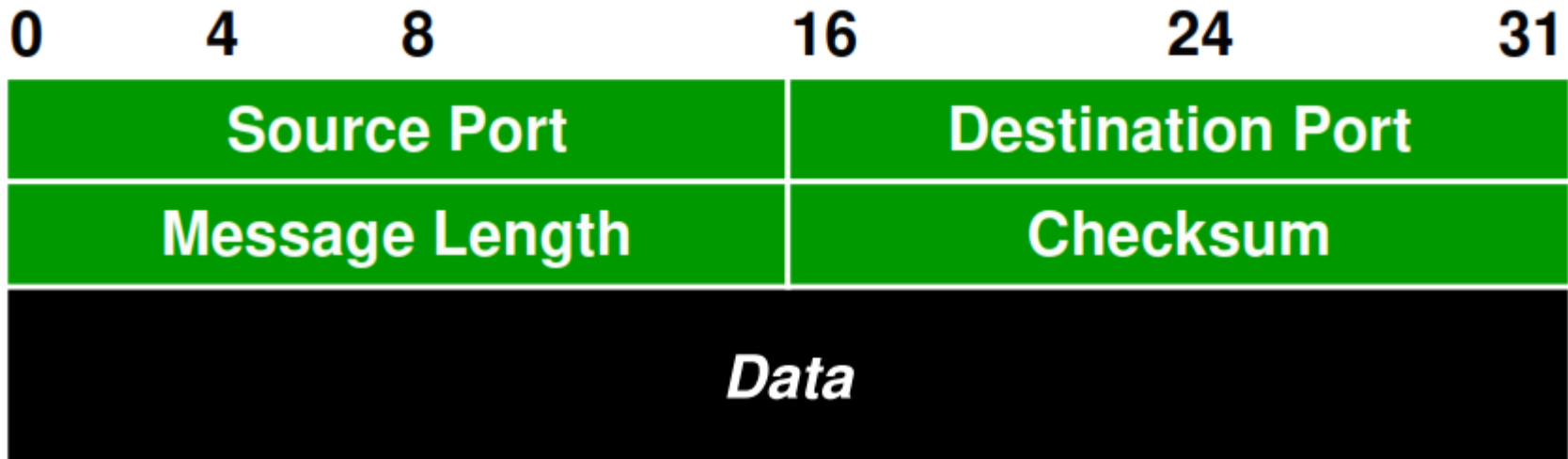
- The User Datagram Protocol (UDP) is called a connectionless, unreliable transport protocol.
- Provide process-to-process communication instead of host-to-host communication.
- It performs very limited error checking.
- Packets may be lost or delivered out of order
- Not buffered -- UDP accepts data and transmits immediately (no buffering before transmission)
- Full duplex -- concurrent transfers can take place in both directions

UDP Header Format



UDP Header details

- UDP Destination Port: identifies destination process
- UDP Source Port: optional -- identifies source process for replies
- Message Length: length of datagram in bytes, including header and data
- Checksum: optional



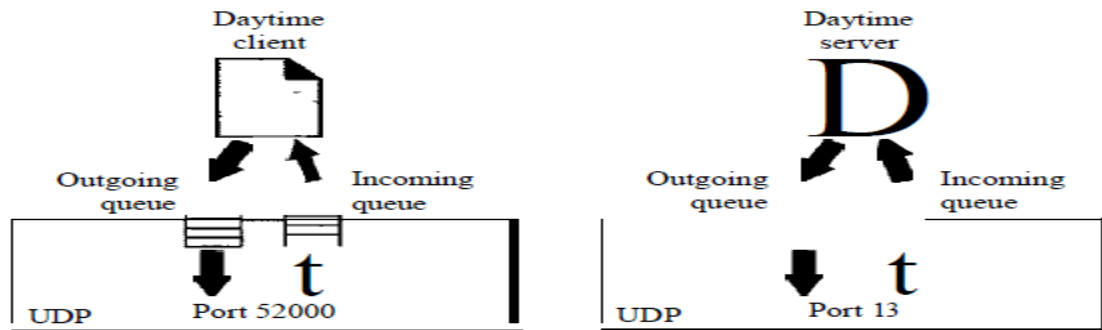
UDP Services

- 1) ***Connectionless Services:*** This means that each user datagram sent by UDP is an independent datagram.
- 2) ***Flow and Error Control:*** There is no error control mechanism in UDP except for the checksum. The lack of flow control and error control means that the process using UDP should provide these mechanisms
- 3) ***Encapsulation and De-capsulation:*** *To* send a message from one process to another, the UDP protocol encapsulates and decapsulates messages in an IP datagram.

UDP Services Continue !

4-Queuing:

- In UDP, queues are associated with ports.
- At the client site, when a process starts, it requests a port number from the operating system. Some implementations create both an incoming and an outgoing queue associated with each process.
- At the server site, the mechanism of creating queues is different. In its simplest form, a server asks for incoming and outgoing queues, using its well-known port, when it starts running. The queues remain open as long as the server is running.



Use of UDP

- Used for the transmission of small amount of data
 - Accuracy is not of prime concern
- Used in video and audio multicasting and broad casting, also used for multimedia transmission
- UDP is suitable for a process with internal flow and error control mechanisms. For example, the Trivial File Transfer Protocol (TFTP) process includes flow and error control.
- UDP is used for management processes such as SNMP
- UDP is used for some route updating protocols such as Routing Information Protocol (RIP) .

Well-known ports used with UDP

<i>Port</i>	<i>Protocol</i>	<i>Description</i>
7	Echo	Echoes a received datagram back to the sender
9	Discard	Discards any datagram that is received
11	Users	Active users
13	Daytime	Returns the date and the time
17	Quote	Returns a quote of the day
19	Chargen	Returns a string of characters
53	Nameserver	Domain Name Service
67	BOOTPs	Server port to download bootstrap information
68	BOOTPc	Client port to download bootstrap information
69	TFTP	Trivial File Transfer Protocol
111	RPC	Remote Procedure Call
123	NTP	Network Time Protocol
161	SNMP	Simple Network Management Protocol
162	SNMP	Simple Network Management Protocol (trap)

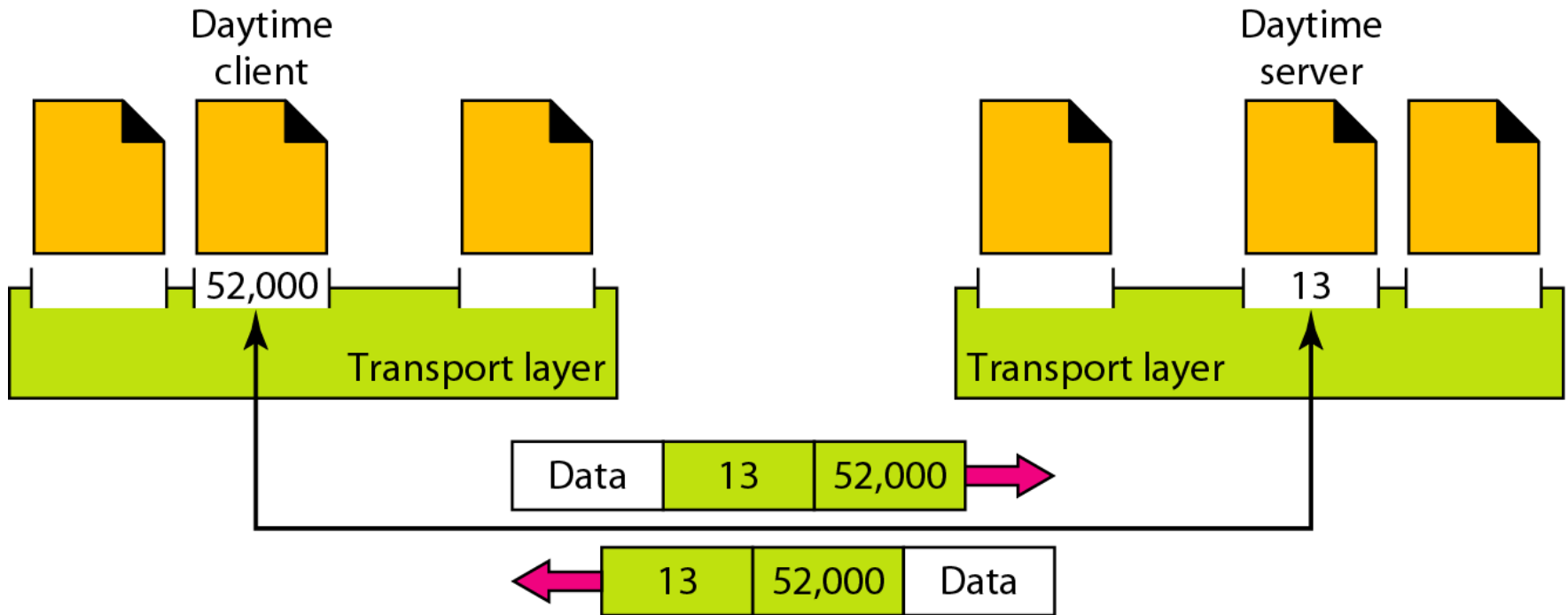
2-Transmission Control Protocol (TCP)

- TCP is a connection oriented protocol; it creates a virtual connection between two TCPs to send data.
- TCP uses flow and error control mechanisms at the transport level.
- TCP is called a *connection-oriented, reliable* transport protocol. It adds connection-oriented and reliability features to the services of IP.
- Used in the accurate transmission of large amount of data
- Slower compared to UDP because of additional error checking being performed

Ports

- TCP requires port numbers on the host and destination for communication
- Standard port numbers have been assigned by the Internet Assigned Number Authority (IANA)
- Standard access for web browsing
 - Ganesan.calstatela.edu
 - Default port of 80 is used in this case
- Non-standard access
 - <http://ganesan.calstatela.edu:5002>
 - The port number 5002 is used in this case to host the above web server

Port numbers

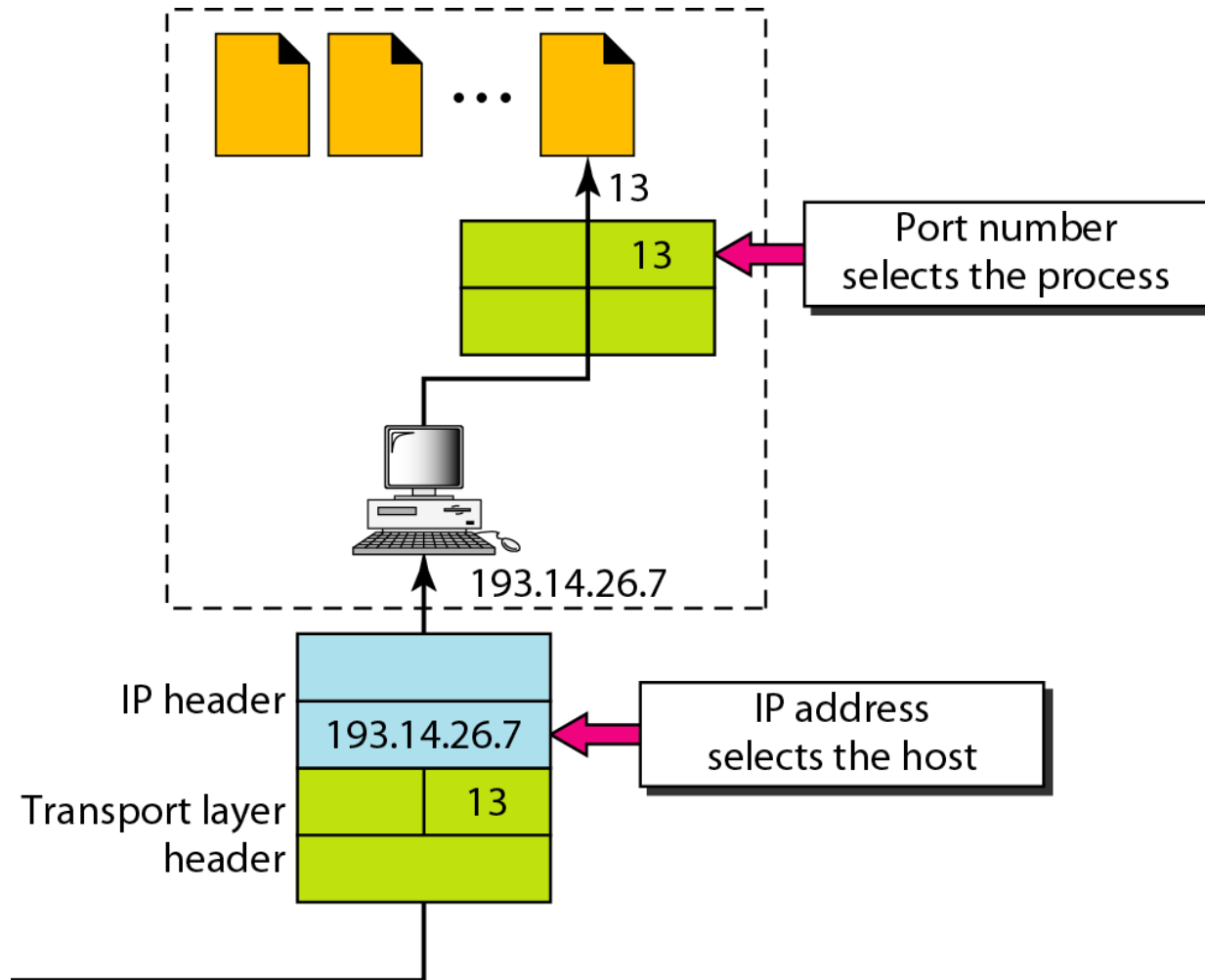


Sample TCP Port Numbers

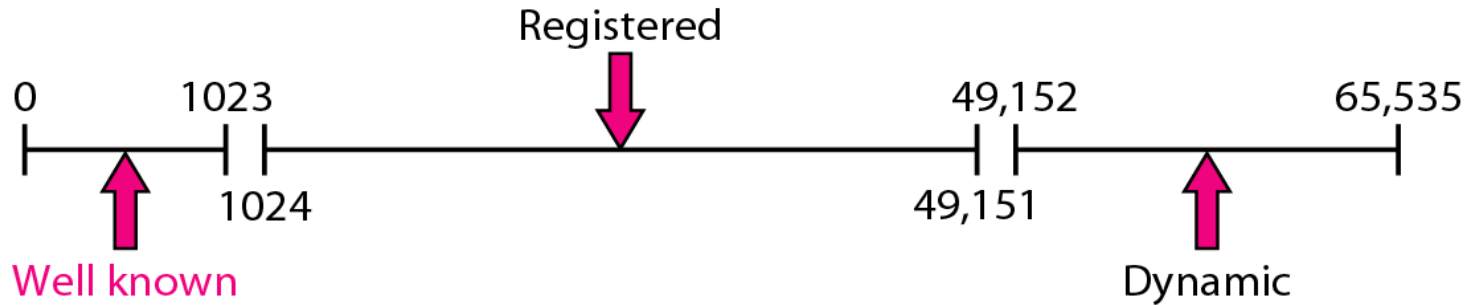
20	FTP Data Channel
21	FTP Control Channel
23	Telnet
80	HTTP on WWW
135	RPC
139	NetBIOS Session Services

Note: There are port numbers applicable to UDP as well.

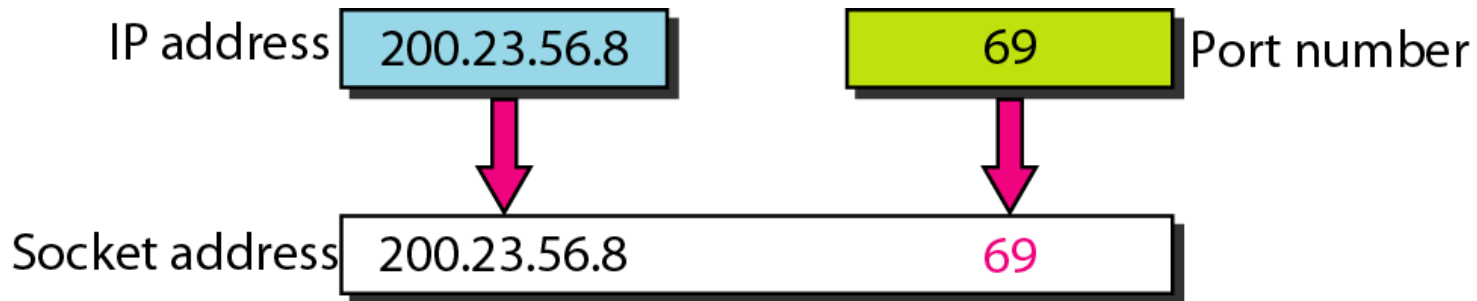
IP addresses versus port numbers



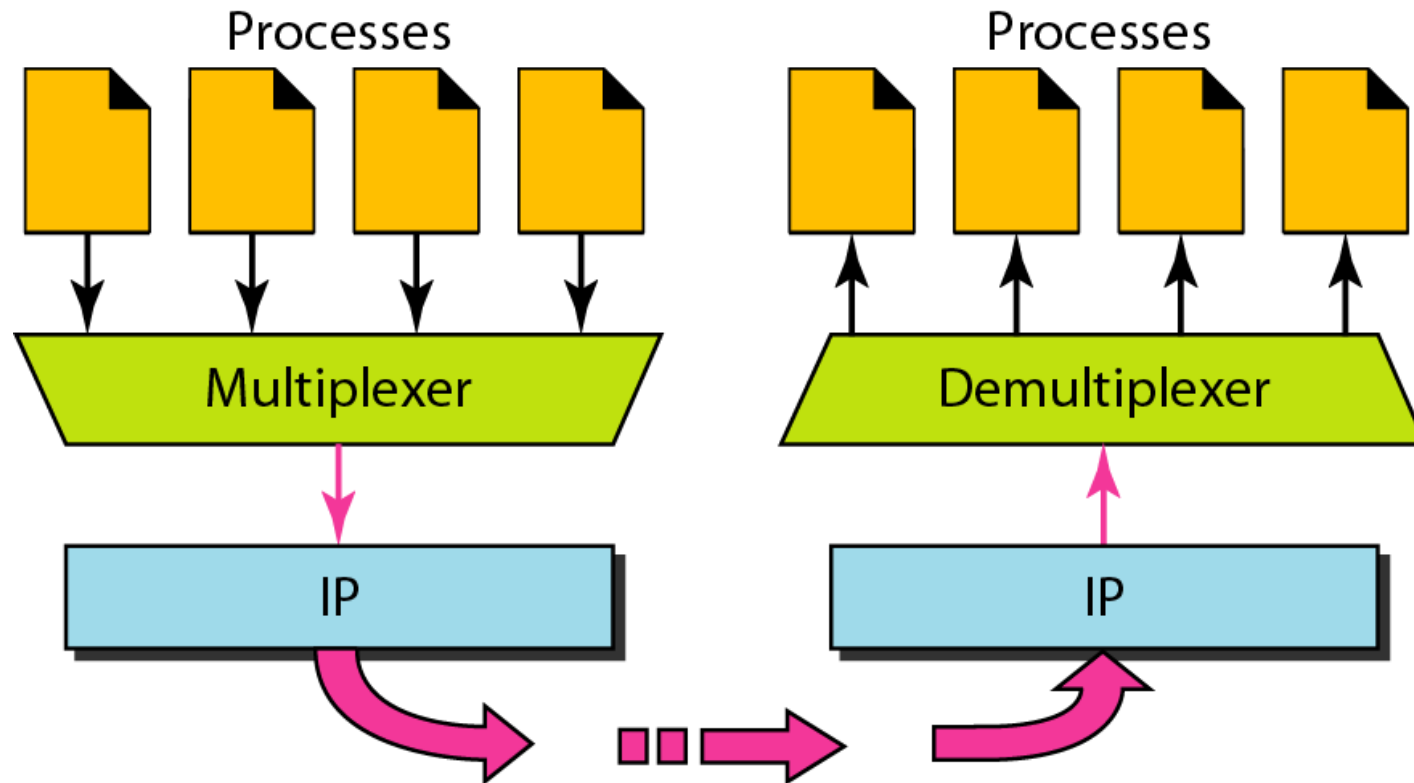
IANA ranges



Socket Address



Multiplexing and demultiplexing



TCP Services:

- A. Process-to-Process Communication*
- B. Stream Delivery Service*
- C. Full-Duplex Communication*
- D. Connection-Oriented Service*
- E. Reliable Service*

TCP Services:

(A-Process-to-Process Communication)

provides process-to-process communication using port numbers.

Table –Below, lists some well-known port numbers used by TCP.

Table 23.2 *Well-known ports used by TCP*

<i>Port</i>	<i>Protocol</i>	<i>Description</i>
7	Echo	Echoes a received datagram back to the sender
9	Discard	Discards any datagram that is received
11	Users	Active users
13	Daytime	Returns the date and the time
17	Quote	Returns a quote of the day
19	Chargen	Returns a string of characters
20	FIP, Data	File Transfer Protocol (data connection)
21	FIP, Control	File Transfer Protocol (control connection)
23	TELNET	Tenninal Network
25	SMTP	Simple Mail Transfer Protocol
53	DNS	Domain Name Server
67	BOOTP	Bootstrap Protocol
79	Finger	Finger
80	HTTP	Hypertext Transfer Protocol

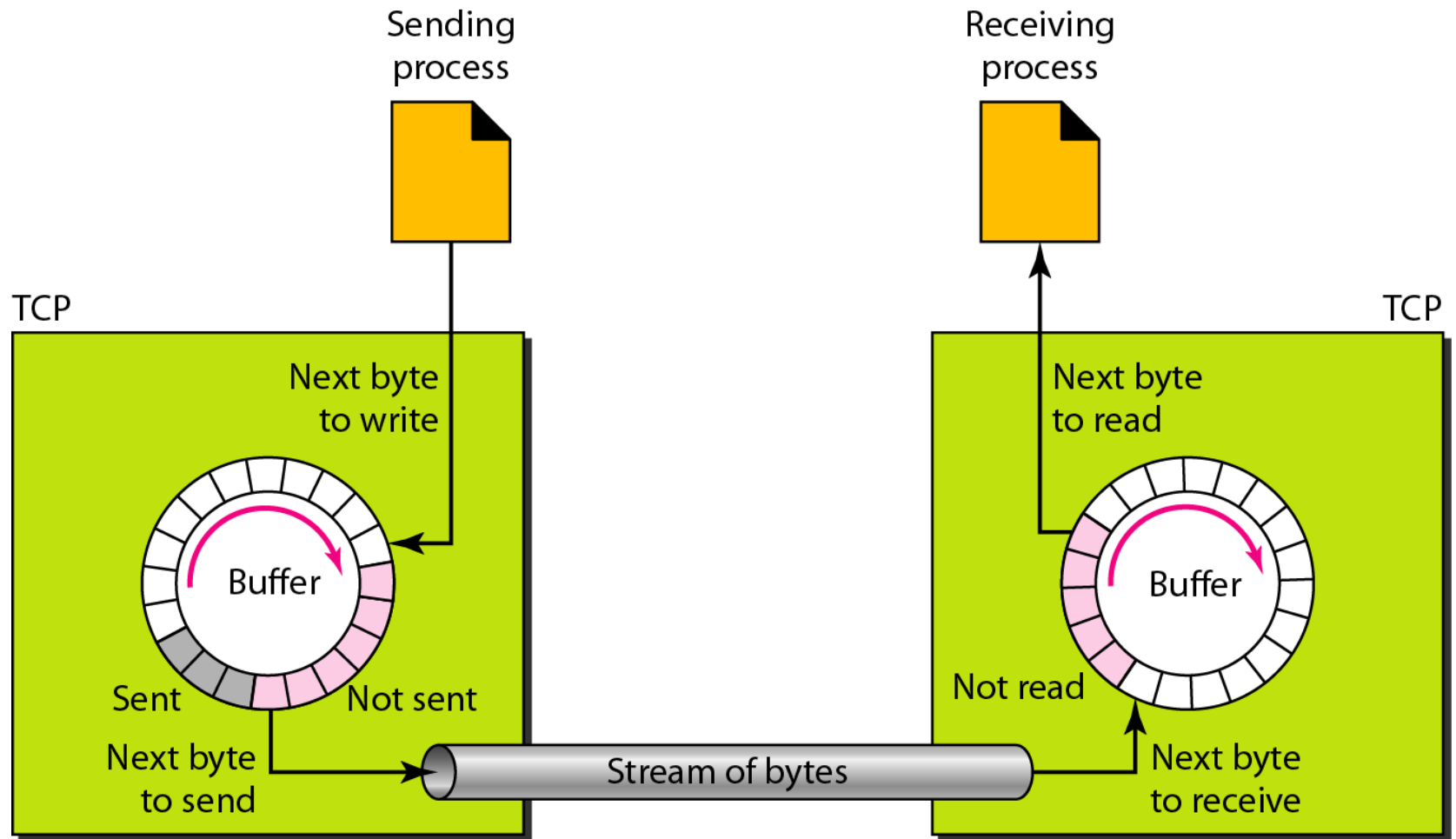
B-Stream Delivery Service

- TCP allows the sending process to deliver data as a stream of bytes and allows the receiving process to obtain data as a stream of bytes.
- TCP creates an environment in which the two processes seem to be connected by an imaginary "tube" that carries their data across the Internet.
- The sending process produces (writes to) the stream of bytes, and the receiving process consumes (reads from) them.

Stream delivery



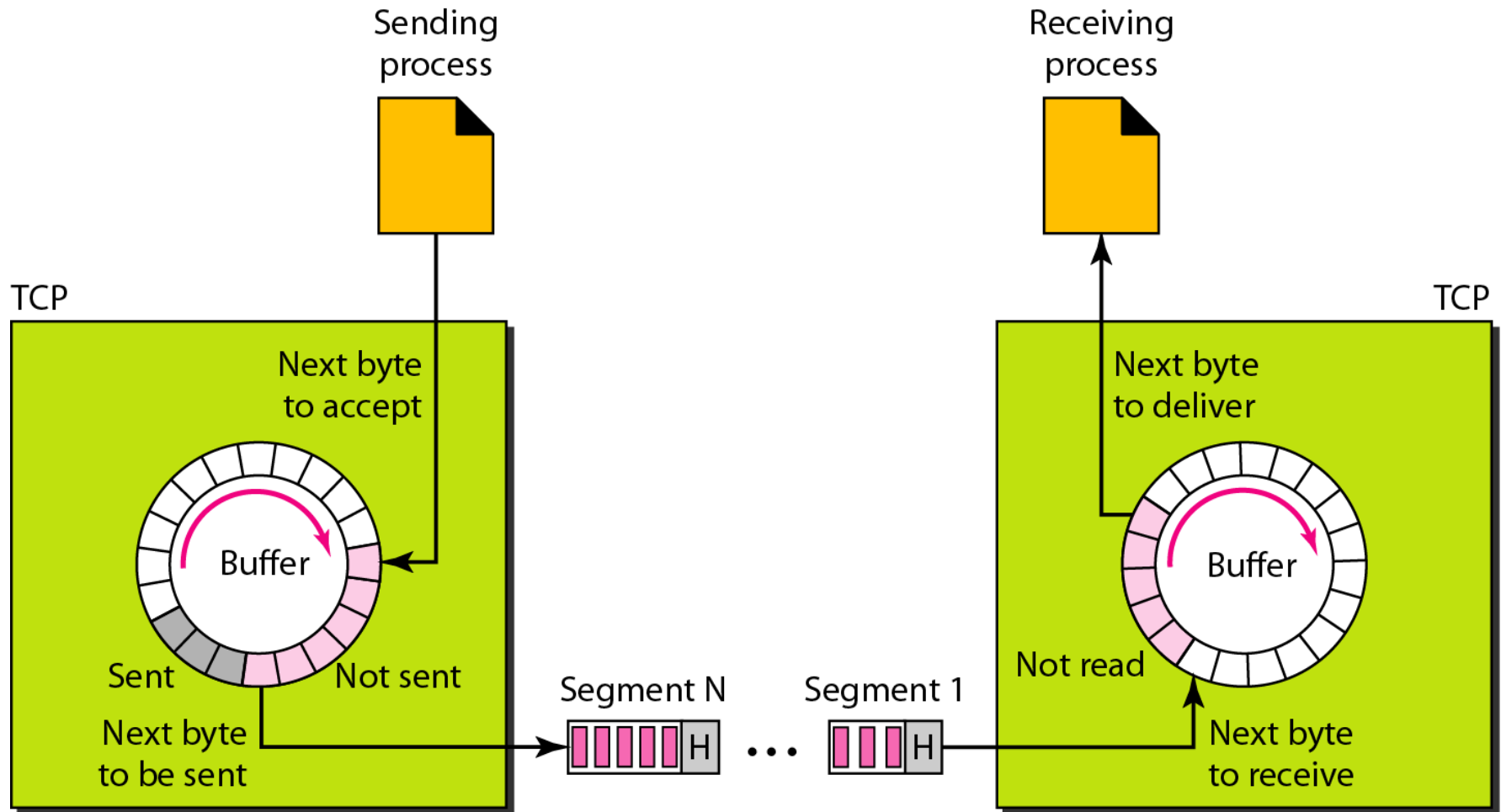
Sending and receiving buffers



C-Full-Duplex Communication

- TCP offers full-duplex service, in which data can flow in both directions at the same time.
- Each TCP then has a sending and receiving buffer, and segments move in both directions

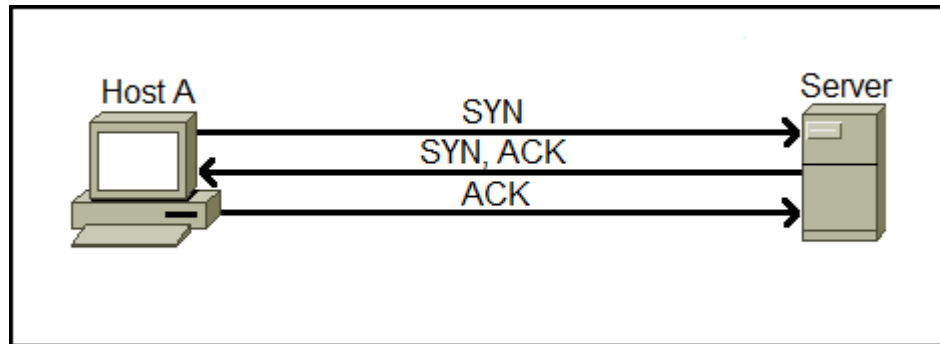
TCP segments



D-Connection-Oriented Service

TCP, is a connection-oriented protocol. When a process at site A wants to send and receive data from another process at site B, the following occurs (This is also called 3- way handshake):

1. The two TCPs establish a connection between them.
2. Data are exchanged in both directions.
3. The connection is terminated.



E-Reliable Service

- TCP is a reliable transport protocol.
- It uses an acknowledgment mechanism to check the safe and sound arrival of data.

TCP Features:

Flow Control:

TCP, unlike UDP, provides flow control. The receiver of the data controls the amount of data that are to be sent by the sender. This is done to prevent the receiver from being overwhelmed with data. The numbering system allows TCP to use a byte-oriented flow control.

Error Control:

To provide reliable service, TCP implements an error control mechanism. Although error control considers a segment as the unit of data for error detection (loss or corrupted segments), error control is byte-oriented.

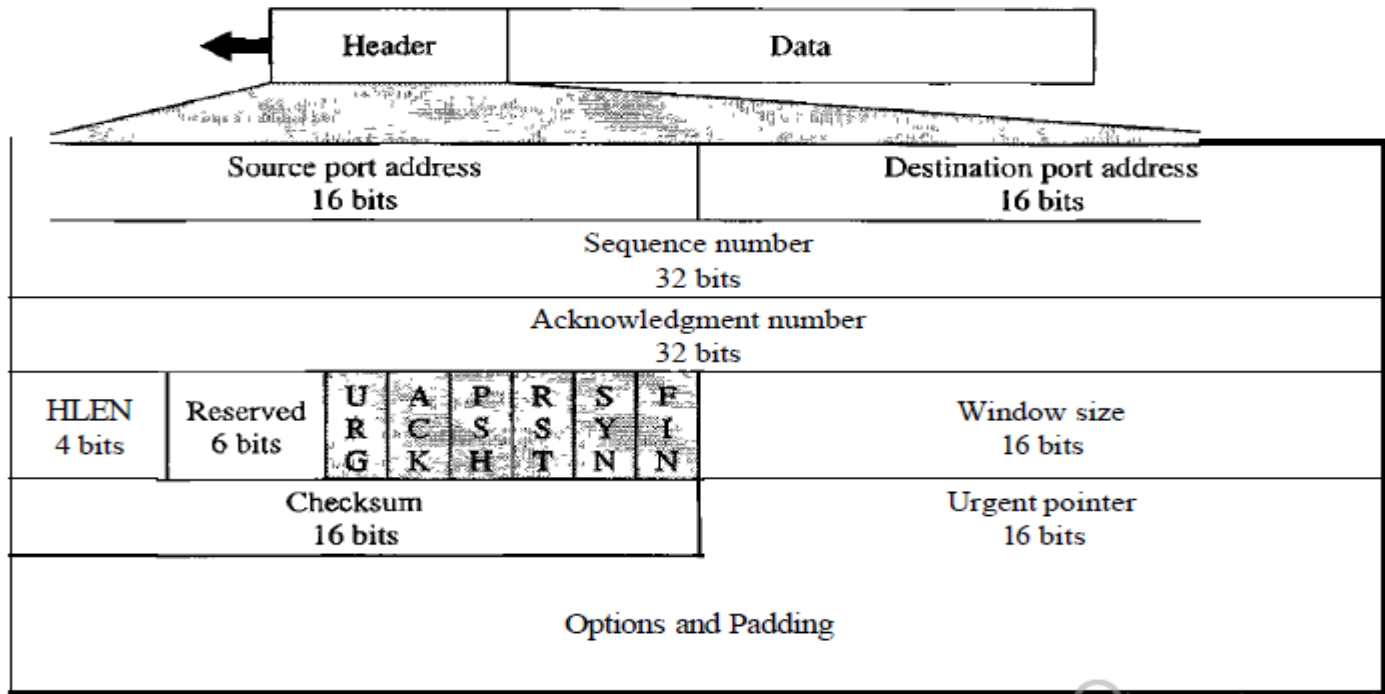
Congestion Control:

TCP, unlike UDP, takes into account congestion in the network. The amount of data sent by a sender is not only controlled by the receiver (flow control), but is also determined by the level of congestion in the network.

TCP Format

Packet in TCP is called a segment.

The format of a segment is shown in Figure



3- Stream Control Transmission Protocol (SCTP)

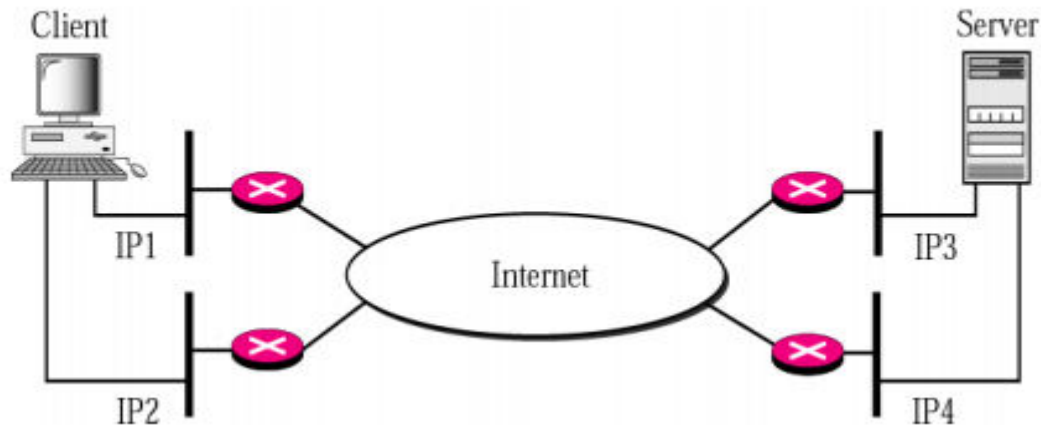
- Stream Control Transmission Protocol (SCTP) is a new reliable, message-oriented transport layer protocol.
- SCTP, is mostly designed for Internet applications that have recently been introduced such as:
 - IUA (ISDN over IP),
 - M2UA and M3UA (telephony signaling)
 - H.248 (media gateway control)
 - H.323 (IP telephony) and
 - SIP (IP telephony)
- SCTP need a more sophisticated service than TCP can provide.
- SCTP provides this enhanced performance and reliability.
- **SCTP** (Stream Control Transmission Protocol) does everything that **TCP** does while adding new features that **TCP** doesn't have.

SCTP Services

- All services provided by TCP are also provided by SCTP except multi-homing and Multi-streaming
- The first feature that **SCTP** adds to the mix is multi-homing. Multi-homing allows the two endpoints of a connection to declare multiple interfaces (IP addresses).

Multi-homing in SCTP

- If the sender or receiver is a multi homed host (connected to more than one physical address with multiple IP addresses in SCTP, the sending and receiving host can define multiple IP addresses in each end for an association.
- In this fault-tolerant approach, when one path fails, another interface can be used for data delivery without interruption.
- This fault-tolerant feature is very helpful when we are sending and receiving a real-time payload such as Internet telephony

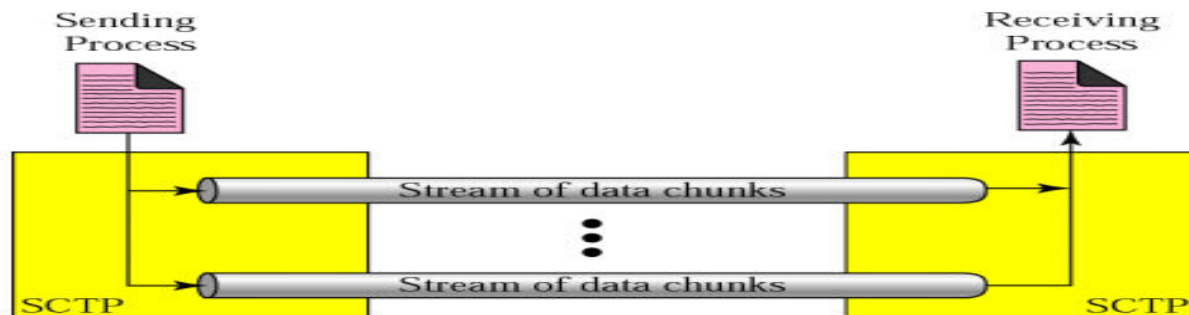


Multi-streaming

- SCTP allows multi-stream service in each connection, which is called association in SCTP terminology. If one of the streams is blocked, the other streams can still deliver their data.
- The idea is similar to multiple lanes on a highway. Each lane can be used for a different type of traffic. For example, one lane can be used for regular traffic, another for car pools.
- If the traffic is blocked for regular vehicles, car pool vehicles can still reach their destinations. Figure below shows the idea of multiple-stream delivery.



Fig 3: Multi-streaming (SCTP Association)



Some SCTP applications

<i>Protocol</i>	<i>Port Number</i>	<i>Description</i>
IUA	9990	ISDN over IP
M2UA	2904	SS7 telephony signaling
M3UA	2905	SS7 telephony signaling
H.248	2945	Media gateway control
H.323	1718, 1719, 1720, 11720	IP telephony
SIP	5060	IP telephony

SCTP Header Format



Differences between TCP, UDP, and SCTP

Attribute	TCP	UDP	SCTP
Reliability	Reliable	Unreliable	Reliable
Connection Management	Connection-oriented	Connectionless	Connection-oriented
Transmission	Byte-oriented	Message-oriented	Message-oriented
Flow Control	Yes	No	Yes
Congestion Control	Yes	No	Yes
Fault Tolerance	No	No	Yes
Data Delivery	Strictly Ordered	Unordered	Partially ordered
Security	Yes	Yes	Improved